

ANIL PRADHAN

BACKEND ENGINEER | AI SYSTEMS BUILDER

Computer Science undergraduate passionate about building scalable backend systems and AI-powered applications. Experienced in FastAPI, LangGraph, RAG pipelines, and cloud deployment. Eager to contribute to impactful products and grow as a software engineer.

anilpradhan9644@gmail.com
+91 8917476908
Bhubaneswar, Odisha, India
linkedin.com/in/anil-pradhan543
github.com/Anil-Pradhan-web
my-portfolio-anil.vercel.app

EDUCATION

ITER, SOA University, Bhubaneswar
B.Tech in Computer Science & Engineering
2024 – 2028 | CGPA: 9.11/10

Odisha Board of Secondary Education
Class XII (PCM with IT)
2022 – 2024 | Percentage: 85%

EXPERIENCE

College Ambassador | Techfest, IIT Bombay

Jun 2026 – Present

- Selected as a Campus Ambassador for Techfest, IIT Bombay, participating in technical and industry-oriented initiatives.
- Contributed to web development and marketing by successfully completing multiple project-based assignments.
- Conducted market research, prepared a business strategy presentation, and qualified for Round 2 of the FlowFinance Growth Strategy & Business Development Challenge.

TECHNICAL PROJECTS

AI Career Mentor | Live Demo | Solo Full-Stack Developer

March 2026 – Present

- Multi-Agent Orchestration & Concurrency:** Orchestrated 7 specialized AI agents using a parallel LangGraph DAG; reduced end-to-end latency by 60% (from 4 mins to 60s) by executing resume parsing and market scraping concurrently.
- System Resilience & Fault Tolerance:** Built failover architecture with custom Circuit Breakers and LLM fallback chains (Groq ⇌ Cerebras ⇌ NVIDIA NIM), preventing request drops during API rate-limit spikes.
- Real-time Bidirectional Streaming & State Machines:** Designed a full-duplex WebSocket protocol with Gemini Multimodal Live API for low-latency audio coaching; built a 7-phase FSM coding interview simulator with Monaco IDE.
- High-Efficiency RAG & Semantic Search:** Optimized ChromaDB RAG using local ONNX embeddings (all-MiniLM-L6-v2) to eliminate API costs; built a keyword-search fallback to prevent OOM container crashes.
- Rate-Limiting, Caching & Telemetry:** Integrated Upstash Redis middleware for sliding-window rate limiting and JWT validation; built a telemetry dashboard tracking API latencies and token cost rollups.
- Data Durability, Dockerization & CI/CD:** Enforced data validation with Pydantic v2, persisting entries to Postgres with PgBouncer pooling; deployed multi-stage Docker builds with a 116 Pytest CI/CD suite.

FastAPI LangGraph ChromaDB ONNX Docker Upstash Redis Next.js 14 Gemini Live

Task Pilot-AI | Live Demo | Integration Lead • Team IdeaForg-E

June 2026 – July 2026

- Solved Developer Context Overload:** Aggregated engineering tasks from 7 sources (Jira, GitHub, Slack, Email, Calendar, Meetings, Incidents) into a unified DB, saving developers ~2 hours daily.
- Recovered Untracked "Hidden" Tasks:** Developed an LLM and rule-based extraction system to parse emails and Slack chats for hidden action items, capturing ~35% of previously untracked tasks.
- Reduced LLM Costs and API Failures:** Built local fallback algorithms to reduce token costs and bypass Groq API rate limits; used fuzzy text-matching for deduplication and local Slack parsing.
- Prevented Database Locking and Workflow Freezes:** Configured SQLite in WAL mode to allow concurrent multi-agent writes without locking; built a self-healing monitor to auto-restart stuck pipeline runs.
- Automated Real-Time Emergency Re-Prioritization:** Created a chat command API allowing developers to inject critical P1 incidents; autonomously re-ran the 6-stage pipeline to update schedules in under 15s.

FastAPI LangGraph SQLite LLMs React Redis

TECHNICAL SKILLS

Languages

Python
Java
JavaScript
SQL

Backend

FastAPI
LangGraph
Pydantic
SQLAlchemy
WebSocket
Prometheus
Sentry
pdfplumber
edge-tts

Frontend

HTML
CSS
Tailwind CSS
Axios
React

Database

Neon Postgres
SQLite
Upstash Redis
ChromaDB

DevOps/Tools

Docker
GitHub Actions
Vercel
Render.com
VS Code
Git

AI/LLM/RAG

Groq Cloud
Cerebras Cloud
NVIDIA NIM
Gemini Live
Tavily
Serper.dev
DuckDuckGo

ACHIEVEMENTS & CERTIFICATIONS

- Google Cloud Arcade Program (2024–25): Completed GCP challenges, earning 75 Arcade Points.
- Solved 200+ problems across GFG and LeetCode, awarded GFG Certificate and T-shirt.
- Certifications: Credentials in Python, AI, and Web Dev from IBM, Cisco, Anthropic, Infosys, Udemy.
- Competitions: Participated in multiple IIT/NIT coding contests and hackathons.
- Hackathons: Built production-grade MVP solutions for Microsoft AI DevDays (\$80K+) and Amazon Nova AI (\$95K+).
- Active participant in Developer Communities (GDG Events) and college tech events.